

Two Narratives: Comparing Domestic and International Framing of Political Speeches

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1 Introduction

Political speeches express political actors’ stances on major policy issues and, thus, they have been the object of analysis in NLP studies. In this report, we investigate the discrepancies in political discourse across domestic and international political settings and how it changes over time, focusing on Italian parliamentary speeches and Italian UN speeches on migration and military-related themes from 2006 to 2022. A multilingual BERT model is fine-tuned on the Manifesto Project dataset and used on the main dataset for a classification task aiming at predicting stances (positive or negative) towards migration and defense policies. To contextualize sentiment results, LDA topic modeling is exploited in order to identify the main topics within migration- and military-related discourse. Finally, a generation task aims at providing summaries of migration and military debates across time and political ideologies.

2 Motivation

Previous NLP literature has shown on that computational methods can be used to study variations in political framing, rhetoric, emotion, and policy preferences across different contexts and platforms, including news, parliamentary debates, and UN speeches [5, 6]. However, less attention has been paid to how the same country frames similar policy issues across different institutional arenas.

This work aims at addressing this gap by comparing political communication across different arenas, namely domestic against international, focusing on Italy. By examining the case of Italian parliamentary speeches in contrast with Italian speeches at the UN, our report aims at addressing how policy stance changes between the two settings and over time from 2006 to 2022. We concentrate on migration and military-related discourse because both are highly salient policy areas in Italian public de-

bate and often polarize political and public opinion. At the same time, they refer to relatively concrete policy domains, which makes them suitable for stance classification and topic modeling, with less ambiguity from metaphorical or abstract language.

3 Data & preprocessing

To fine-tune the multilingual BERT model, we use selected quasi-sentences from the Manifesto Project dataset [8], which collects party manifestos from 67 countries from 1945 onward and annotates parties’ policy positions. Since it provides labeled political language, we select the categories related to migration and military defense and recode them into positive and negative stance labels.

The analysis relies on two main corpora. The first is ItaParlCorpus[2], containing Italian Chamber of Deputies speeches from 2006 to 2022 in the original language. We retain key metadata, including party family, to enable partisan analysis. The second corpus is built by manually collecting Italy’s speeches from the United Nations General Debate Corpus[1][7] over the same period. These speeches are in English and represent the official Italian government position rather than partisan debate.

Texts are lightly normalized to make the corpora comparable while keeping preprocessing computationally feasible. Italian speeches are segmented into sentences using spaCy’s `it_core_news_sm` pipeline[3]. We then filter sentences through an extensive keyword list using word-boundary regex, a compromise between plain substring search and full lemmatization, which was impractical at this scale. Since political language is often ambiguous, the resulting data still contains some noise. We also experimented with a bi-encoder expansion followed by cross-encoder re-ranking, but excluded this step from the final pipeline because the marginal improvement in data quality did not justify the additional computational cost.

Given the volume of retained data, we sample 100 speeches per year $\times$ party-family group, ranked by keyword hits. We use speeches, rather than isolated sentences, as the unit of selection because sentence-level excerpts often lack the discourse context needed for coherent topic modeling and summarization.

4 Methodology & Results

4.1 BERT Classification

For the supervised classification task, we use multilingual BERT [4], because the speeches analyzed are both in English and Italian. This model was pre-trained on Wikipedia in multiple languages, which gives a good basis for further adaptation to the political domain.

The model is fine-tuned on selected quasi-sentences from the Manifesto Project dataset. The predicted binary label represents the direction of the policy stance (positive or negative), and is constructed from several topics codes relevant for this project. Before training, the dataset is balanced manually in order to stop the model from overfitting to specific language usage. Specifically, all negative examples and all Italian positive examples are kept, because they are relatively scarce, while English positive examples are sampled to reach the target dataset size and to preserve the country distribution observed among English negative examples. The final fine-tuning dataset contains 5,000 observations. The split to train and developmental datasets is stratified by both language and label, so that the positive/negative ratio is preserved within each language.

The model reached approximately 0.81 macro-F1 on the development set. As shown in Tables 1 and 2, performance is relatively balanced across the two classes: the F1-score is 0.79 and 0.82 for the negative and positive classes, respectively. This suggests that the model does not strongly favor one class over the other and is able to identify both negative and positive policy-stance statements with comparable accuracy.

After fine-tuning, the model is applied to four downstream datasets. The overall results are reported in Table 3 in Appendix. Parliamentary speeches are classified as predominantly negative: 54% of migration-related and 69% of military-related parliamentary sentences receive a negative label. In contrast, UN speeches are classified as predominantly positive, with 72% positive predic-

tions for migration-related sentences and 60% for military-related sentences. This difference may reflect institutional context: UN speeches are delivered in a more diplomatic international setting, while parliamentary debates often include more domestic political contestation. At the same time, part of the pattern may also come from language and domain effects.

As a further step, we analyze how the predicted sentiment changes across three periods: 2006–2013, 2014–2017, and 2018–2022. The detailed results are visualized in Figure 1. The most interesting dynamics appear in two cases. First, migration-related parliamentary discourse gradually moves toward more positive sentiment: from 43% in 2006–2013 to 49% in 2018–2022. This may indicate that migration debates remain contested, but there emerges a shift from crisis-oriented framing toward more pragmatic discussions of integration. Second, military-related UN discourse fluctuates strongly: positive sentiment rises from 58% to 72% in the middle period, and then falls to 55% in 2018–2022. This pattern may reflect changes in the international security context, including the Crimea 2014 crisis, and later the shift of attention toward COVID-19 and cooperation-oriented framing.

Finally, to add extra validity to the model results, we use an independent annotator to classify a random subsample of 100 observations. Overall, the comparison results are reasonably encouraging: 72% of the model predictions match the human labels. Agreement is higher for texts predicted as positive, as well as for English ones. Future research could improve the reliability of the validation by expanding the supervised fine-tuning dataset, adding more Italian and negative training examples, and involving multiple annotators to measure inter-annotator agreement.

4.2 LDA Topic Modeling

To further interpret results of the classification task, an unsupervised topic modeling task is performed using LDA to identify the main themes in migration- and military-related sentences and assess whether changes in predicted stance are associated to changes in learnt topic prevalence.

LDA is applied separately to the four datasets using the same pipeline. Texts are preprocessed with spaCy, and custom stopwords are added to remove frequent procedural and domain-specific terms typical of political speeches. Since LDA relies on

word counts, documents are represented as Bag-of-Words. Finally, very rare and very common tokens are filtered from the dictionary to reduce noise and improve interpretability.

The choice of the number of topics, denoted by K , is made by comparing coherence scores and manually inspecting topic words for different values of K . In particular, C_v is used as a metric of coherence, since it is the most correlated to human judgments. The number of topics, coherence scores and diversity scores for each model are reported in Table 4. Overall, all models show relatively high topic coherence and diversity scores with the partial exception for the UN military corpus. As shown by the topic words Tables 5, 6, 7, 8, this model exhibits more lexical overlap across topics, suggesting that military-related political discourse may rely on a more shared diplomatic vocabulary and, hence, less sharp themes distinctions detected by the model. Finally, it is worth noting that, as expected in topic modeling, some themes are more generic compared to other topics, which capture a clearer angle on migration or military issues.

As noted above, migration UN speeches are classified as predominantly positive compared to parliamentary speeches. These differences in predicted stance are consistent with the themes identified by the LDA models. In the UN corpus, themes are centered on international migration, humanitarian challenges, refugee support and migrants crises. By contrast, in a domestic setting as the Italian parliament, the corpus includes topics related to immigration law, borders, asylum, economy and domestic political debate. This suggests that the more positive stance detected in UN speeches may partly reflect migration framed as a humanitarian and cooperation issue, whereas parliamentary discourse frames migration also through domestic regulation, control, and political contestation. Instead, in the military corpus, as expected, the parliamentary topics show greater emphasis on domestic themes, such as armed forces and defense legislation, whereas the UN topics focus more on Italy’s international commitments and international war-related crises.

As a further step of the analysis, we aggregate topic distributions by year periods as defined above for each dataset. The results are exhibited in Figures 2, 3, 4, 5. Interestingly, among parliamentary speeches, it can be observed an increase in two topics, "European and international migration flows"

and "Borders and asylum" between 2006-2013 and 2014-2017, with the former having a similar average topic probability level also in 2018-2022. This pattern is consistent with Italy’s migration history, with migratory flows reaching peaks in 2014 and 2016 [11]. Moreover, this finding may better contextualize the predicted stance in the classification analysis. Regarding military-related speeches, the topic shifts are less directly linked to predicted stance, but may still be informative. In 2014-2017, average topic probability increases for "International security and military missions" in the UN corpus and for "European defense and Ukraine" in the parliamentary corpus. This coincides with the period in which military-related UN speeches receive the highest share of positive predictions, possibly reflecting a more cooperation-oriented framing of international security. By contrast, in 2018-2022, the rise of the "Ukraine and Libya crises" topic in the UN corpus occurs alongside a decline in positive stance, suggesting a partial shift toward more crisis-centered discourse.

Lastly, to validate the model qualitatively, we ask an annotator to assign a topic label to a random subsample of 25 documents for each dataset, for a total of 100 documents. As a measure of human-model topic agreement, we compute top- k match rates, defined as the proportion of manually annotated documents where the human topic label is included among the model’s k most probable topics, for $k = 1, 2, 3$. The results are shown in Table 9. Overall, human-model agreement is satisfactory, especially when considering the model’s three most probable topics. The main exception is represented by the UN military dataset, which shows a lower top-1 match rate, consistently with the stronger lexical overlap across topics discussed above.

5 Generation

5.1 Translation

Finally, we implemented a summarization task in order to provide an interpretative layer to the preceding analysis. To prepare the data for the task we first had to translate the speeches in order to have a single language. We used the model OPUS-MT it-en[10], preferring to translate from Italian to English as it allows a broader choice of models for the generation task and enables use of the strongest summarization architectures which are predominantly English-trained. The translation was done

at chunk-level as a technical necessity since the model allows for a maximum of 512 tokens.

5.2 Summarization

For the summarization task, we first tried summarizing at the sentence level, treating the pre-filtered sentences as the extraction stage output and applying the generation model directly on top of them. However, this produced incoherent summaries because single sentences often lacked the context needed to understand the full meaning. We therefore shifted to a speech-level hierarchical summarization approach, which preserves more context by summarizing larger text units step by step.

We used BART-large-CNN[9], a model suited to summarizing English news-style text by both selecting key information and rewriting it concisely. Since BART has a 1,024-token input limit, full speeches could not be summarized directly. The text was therefore reduced in three stages: first by chunk, then by speech, and finally by group, defined as $\text{source} \times \text{topic} \times \text{party family} \times \text{year}$, which is our unit of interest. We also implemented batching optimization to substantially reduce the estimated runtime.

A limitation of our approach lies in the hierarchical summarization strategy: while this technique is necessary to handle the volume of the data, it also introduces its own distortions. Reducing potentially hundreds of speeches into a single paragraph inevitably flattens the discourse. Indeed, minority positions within a party group, outlier speeches, and temporal shifts across the year are all absorbed into a single compressed representation. This also manifests at the linguistic level: hierarchical summarization is known to struggle with preserving logical connectives, meaning the final summaries read as collections of statements rather than coherent arguments. In conclusion, this implies that the summaries are best understood as representative snapshots rather than exhaustive accounts, and should be interpreted with awareness of these limitations.

The summarization pipeline produces a structured, human-readable layer over the filtered corpora. While the output was not directly used as input for subsequent modelling, indeed the sentiment analysis and topic modelling tasks operated on the raw filtered corpora, it offers a qualitative interpretative complement to those results. A researcher reading the topic distributions or sentiment scores

for a given party family in a given year can consult the corresponding summary to ground those findings in the actual content of the speeches. That said, the group-level structure of the output, organised by topic, source, party family, and year, makes it a natural candidate for future work; whether as a document-level feature, a qualitative validation tool, or an input to retrieval-based pipelines.

6 Conclusion

6.1 Limitations and future research

Nonetheless, it is worth noting that our analysis has some limitations. First, in the fine-tuning dataset, the representation of Italian corpus and negative labels is lower and fine-tuning could be improved using labeled speeches data as opposed to Mani-festo's. We acknowledge that UN data is limited especially for earlier time periods considered. Hence, expanding the international political setting dataset may help deepen the analysis. Finally, the absence of a reference gold standard precluded any systematic evaluation of summary quality, leaving manual spot-checking as the only validation method; combined with the register mismatch between BART's training data and translated parliamentary speech, the absolute quality of the summaries cannot be formally guaranteed.

6.2 Contributions

In conclusion, our work presents the use of computational and NLP methods to show discrepancies in Italian political discourse on migration and defense policies across institutional settings, both in terms of political stance and in terms of thematic framings. In particular, we highlighted the use of BERT for classification and LDA topic modeling to analyze co-trends by year periods. Beyond the specific findings, the report proposes a reproducible pipeline that can be extended to other countries, policy domains, or institutional arenas to compare how political actors adapt their stance and framing across contexts.

The generative summarization task adds a complementary interpretative layer, making large-scale discourse more accessible while attempting to define patterns across the content of the speeches. More innovatively, the summaries can serve as a reusable analytical layer for further research, supporting future comparisons across parties, years, and institutional settings.

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A Appendix

Table 1: Confusion matrix on the development set

	Pred. neg.	Pred. pos.
True neg.	362	94
True pos.	97	447

Table 2: Classification performance on the development set

Class	Prec.	Rec.	F1	Support
Negative	0.79	0.79	0.79	456
Positive	0.83	0.82	0.82	544
Accuracy			0.81	1000

Table 3: Overall BERT predictions across downstream datasets

Dataset	Negative	Positive
Migration–Parliament	53.8%	46.2%
Military–Parliament	69.2%	30.8%
Migration–UN	28.1%	71.9%
Military–UN	39.7%	60.3%

Table 4: Number of topics and coherence scores for each LDA model

Dataset	K	C_v	Diversity
Migration–Parliament	7	0.405	0.881
Military–Parliament	4	0.420	0.813
Migration–UN	5	0.454	0.833
Military–UN	4	0.457	0.667

Table 5: Topic words for the Migration–Parliament LDA model

Topic label	Topic words
Immigration law	espulsione, legge, norma, irregolare, immigrazione, diritto, decreto, reato, rimpatrio, permesso_soggiorno
European and international migration flows	italia, europa, europeo, migrante, flusso_migratore, umano, paese, sbarco, libia, accordo
Labour market and economy	cassa_integrazione, lavoratore, impresa, euro, azienda, fondo, settore, straordinario, dipendente
Borders and asylum	confine, territorio, rifugiato, richiedente_asilo, accoglienza, immigrato, migrante, centro_accoglienza, frontiera, sicurezza
General immigration debate	immigrato, italiano, parlare, immigrazione, italia, pensare, problema, clandestino, sapere
Social integration, inclusion, cohesion	integrazione, sociale, inclusione, europeo, politica, economico, coesione_sociale, sviluppo, inclusione_sociale, percorso
Reception system, services, training	ricollocazione, formazione, disabilità, migrazione, servizio, piano, corso, sistema_accoglienza, attività

Table 6: Topic words for the Migration–UN LDA model

Topic label	Topic words
International migration and integration	migration, international, need, risk, ukraine, integration, action, italy, european, phenomenon
Humanitarian challenges	migration, life, challenge, human, fight, trafficker, flow, mediterranean, community, dignity
Refugee crisis and support	refugee, migration, work, country, crisis, italy, migrant, development, support, humanitarian, mediterranean
Italy and migrants	migration, migrant, italy, thousand, immigrant, refugee, global, country_origin, operation, people, ship

Table 7: Topic words for the Military–Parliament LDA model

Topic label	Topic words
International missions/conflicts	militare, missione, libia, afghanistan, guerra, internazionale, iraq, siria, pace, libano, intervento
Defense and security legislation	militare, articolo, difesa, legge, sicurezza, personale, decreto, forza_armato, servizio, polizia, ministero
Legal-constitutional/institutional procedure	commissione, chiedere, costituzionale, giustizia, commissione_affare, intervento, difesa, cittadino, tutela, pubblico
Domestic armed forces	militare, esercito, carabinieri, polizia, guardia_finanza, forza, ordine, italiano, forza_armato
European defense and Ukraine	difesa, italia, europeo, ucraina, sicurezza, militare, politico, diritto, europa, rispetto

Table 8: Topic words for the Military–UN LDA model

Topic label	Topic words
Ukraine and Libya crises	ukraine, country, italy, libya, security, military, syria, force, support, mission, peace, crisis
Lebanon, Africa, Middle East crises	lebanon, africa, ukraine, libya, afghanistan, middle_east, mediterranean, interim_force, peacekeeping_operation, contributor
Italy’s commitment	italy, libya, support, continue, country, commitment, work, lebanon, situation, contribute, syria, afghanistan
International security and military missions	international, force, security, mission, country, military, commitment, crisis, support, region, humanitarian, afghanistan

Table 9: Manual validation: top- k human–model topic match rates

Dataset	Top-1	Top-2	Top-3
Migration–Parliament	0.40	0.52	0.64
Military–Parliament	0.44	0.64	0.88
Migration–UN	0.44	0.68	0.80
Military–UN	0.08	0.56	0.84

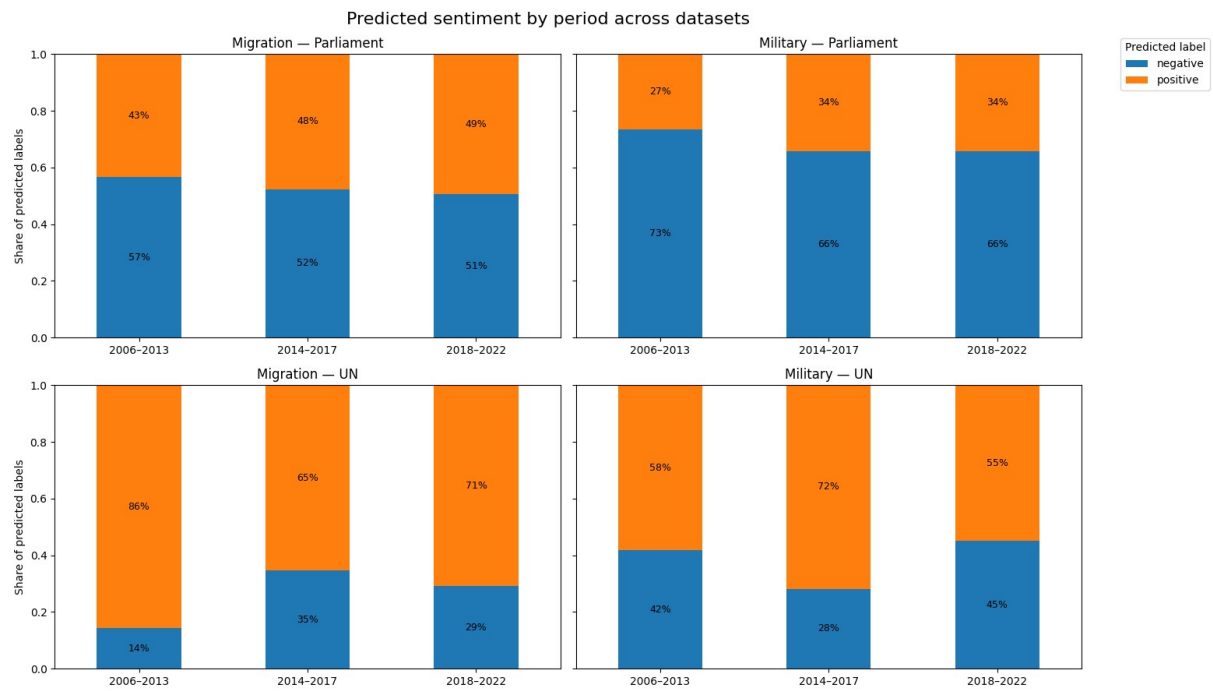


Figure 1: Predicted sentiment by period

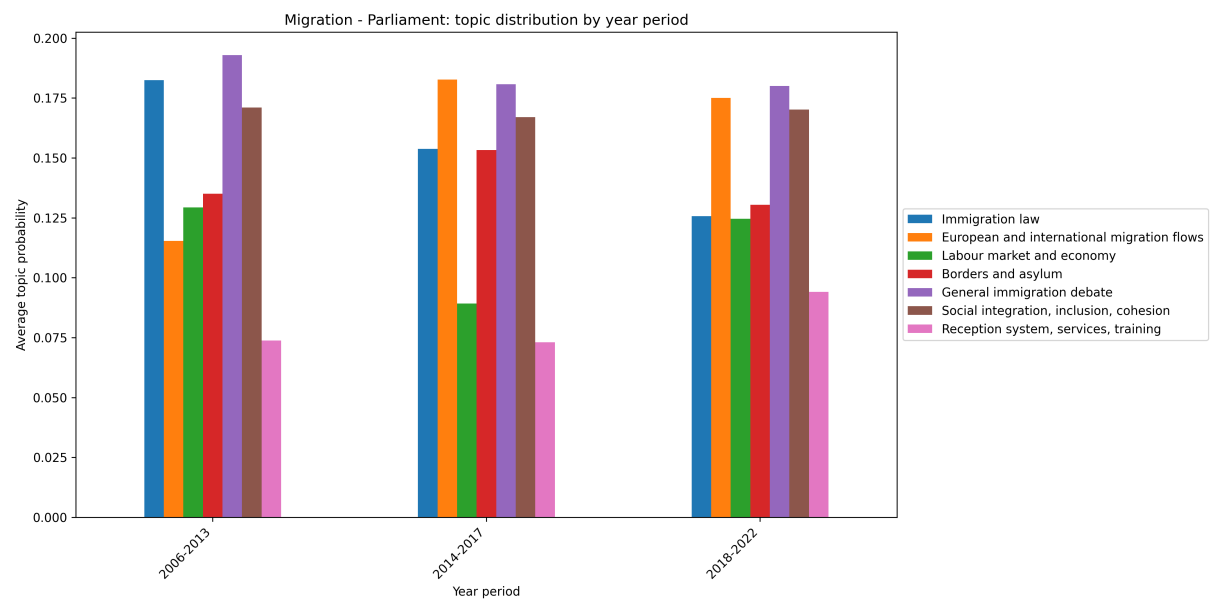


Figure 2: Migration – Parliament: topic distribution by year period

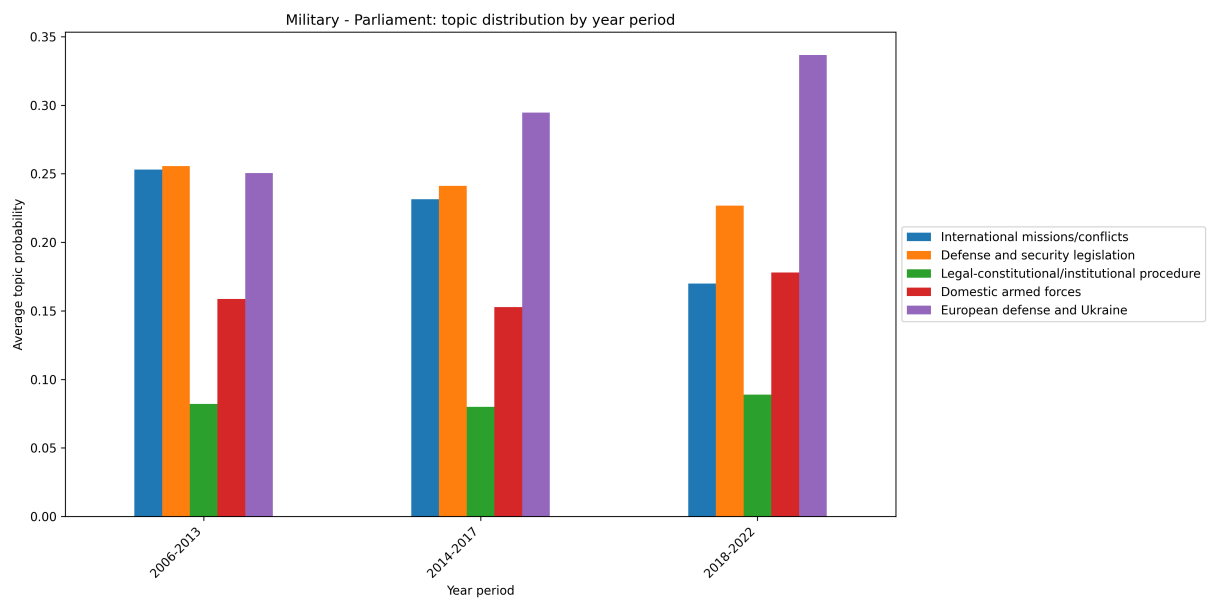


Figure 3: Military – Parliament: topic distribution by year period

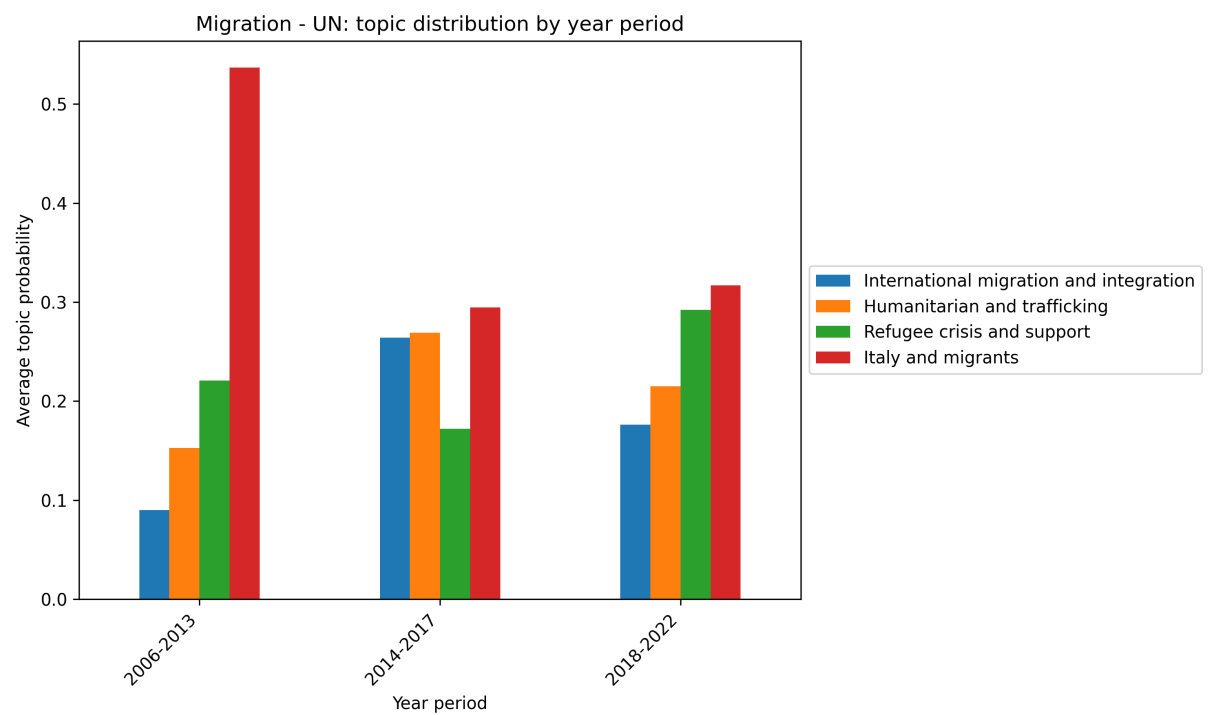


Figure 4: Migration – UN: topic distribution by year period

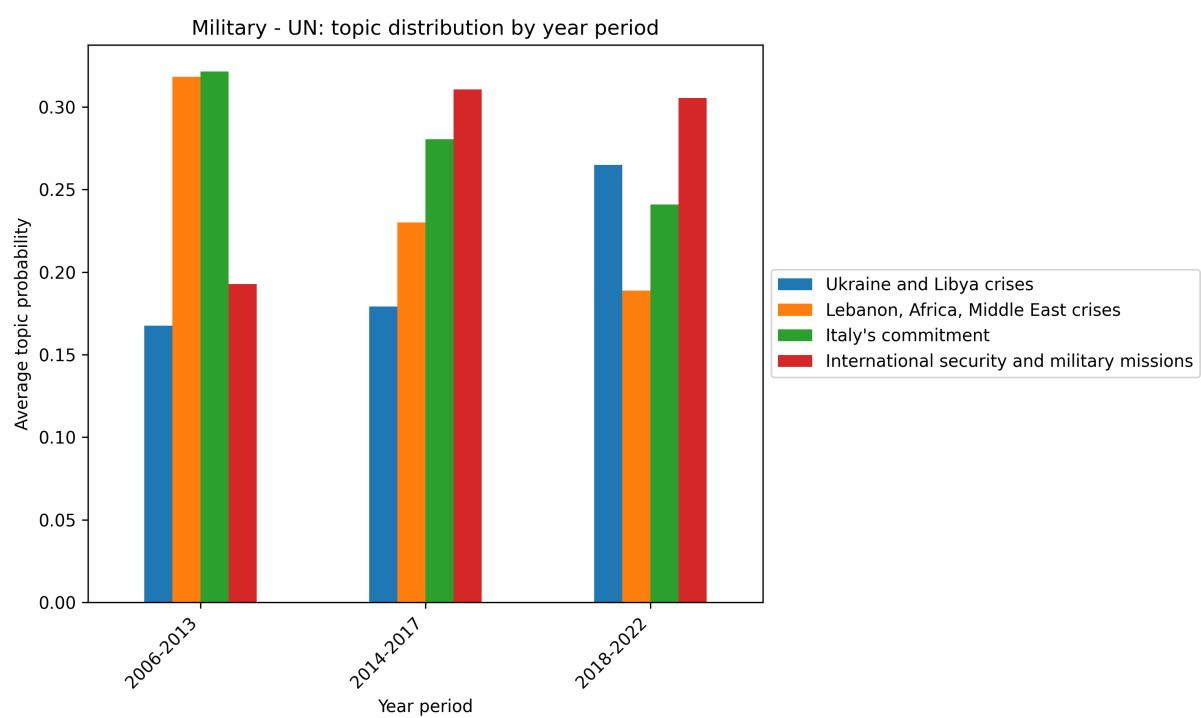


Figure 5: Military – UN: topic distribution by year period